3SAT $\longrightarrow$ zero-finding with constant Jacobian

CLAIM

For every 3CNF formula ϕ with n variables and m clauses, one can construct a degree-7 polynomial map $K_\phi : \mathbb{R}^{3n+3m} \to \mathbb{R}^{3n+3m}$ such that

$$K_\phi^{-1}(0) \neq \varnothing \iff \phi \in \mathrm{SAT}, \qquad \det JK_\phi \equiv (-3456)^{n+m}.$$

Proof idea. Fix a polynomial map $H : \mathbb{R}^3 \to \mathbb{R}^3$ with $\det JH \equiv -3456$, with exactly three preimages of zero, $H^{-1}(0) = \{r_0, r_+, r_-\}$, and with some omitted target $\Delta \notin \mathrm{im}(H)$. The decoder $\beta(x, y, z) = x^2$ reads r_0 as 0 and $r_\pm$ as 1.

Given ϕ, give each variable X_i a block $w_i \in \mathbb{R}^3$ and include $H(w_i)$ in the output. Therefore every zero forces $X_i = \beta(w_i) \in \{0, 1\}$. For each clause C_j, let s_j be its usual polynomial Boolean value, so $s_j = 1$ exactly when C_j is true. Give the clause a fresh block v_j and output $T_{s_j}(v_j) = H(v_j) - (1 - s_j)\Delta$. Since H reaches 0 but never reaches Δ,

$$\exists v_j : \; T_{s_j}(v_j) = 0 \iff s_j = 1.$$

Stack all variable and clause blocks to obtain K_ϕ. It has a zero exactly when every clause is true. Moreover, each T_{s_j} differs from H only by an output translation, so JK_ϕ is block lower triangular with copies of JH on the diagonal. Hence $\det JK_\phi = (-3456)^{n+m}$, while $\deg K_\phi \leq 7$.